

EDUCATION

Indian Institute of Technology Madras, India
Engineering Physics B.Tech. (Hons.) | CGPA: 9.09 / 10.00

2022 – 2026

Delhi Public School, R.K. Puram, India
Class 12 - Central Board of Secondary Education (CBSE) | Grade: 97.4 % (Top 3 in PCM Graduating Class)

2022

RESEARCH EXPERIENCE

Undergraduate Researcher || GEANT SIMULATION, ELECTRONICS & DAQ
PI: Dr. David Koilpillai

Aug 2025 – Present

Space Lab, IIT Madras, India

- Contributed to **Space-based Proton Electron Energy Detector (SPEED)** payload for IITMSAT-ISRO satellite launch.
- Modernised Geant4 simulation framework** and performed detector simulations to model detector efficiency, and response to cosmic muons and P-32 obtaining energy spectrum for energy calibration.
- Executed Data Acquisition (DAQ) workflows** using MC^2 Analyzer for signal readout and monitoring signal with digitizers.
- Co-led C firmware development** for real-time charged-particle flux monitoring, resolving timer configuration issues, extending High Voltage Ramping Timer limits, and debugging ADC-to-energy mapping for accurate data accumulation.

Research in High-Energy Physics (CMS Experiment) || GRAPH ML & JET SUBSTRUCTURE

Summer Research Intern (RPIT Fellowship)

Jun 2025 – Aug 2025

PI: Dr. Matthew Nguyen, Dr. Shamik Ghosh

LLR, École Polytechnique, France

- Engineered **FusionNet**, a novel graph neural network (GNN) track classifier with auxiliary b-jet tagging, published in a CMS DPS; **outperformed Boosted Decision Trees** by ~5-15% in high signal-efficiency regions.
- Leading development** of a variational graph autoencoder (VGAE)-based contrastive clustering algorithm to analyze jet substructure in pp collisions using physics-informed constraints for **meaningful representations**.
- Diagnosed model limitations** via performance plots and score distributions; identified irreducible errors from track-matching inefficiencies, guiding refinements in data representation and architecture.

Young Research Fellow (Funded Undergraduate Research)

Jun 2024 – May 2025

PI: Dr. Prabhat Pujahari

EHEP Lab, IIT Madras, India

- Explored **track classification** from b-hadron decays in pp and PbPb collisions in CMS at $\sqrt{s} = 5.02$ TeV, investigating the **dead cone effect** and **b-flavor physics** using a GNN-based deep learning framework.
- Developed an **end-to-end framework** for graph-based track classification, including dataset construction, model training with PyTorch & torch-geometric, evaluation of physics-motivated performance metrics.
- Engineered a preprocessing pipeline** with uproot and awkward-array to handle 50GB of ROOT data, apply physics cuts, match events-jets-tracks via pandas, and serialize into HDF5 batches for scalable training.

Undergraduate Researcher || FIRMWARE DEVELOPMENT

Nov 2024 – May 2025

PI: Dr. Prafulla Kumar Behera

Detector Development Lab, IIT Madras, India

- Worked on MALTA-to-FPGA communication firmware for next-gen readout chain for the ATLAS experiment.
- Executed end-to-end FPGA workflow** on ALINX board: simulation, synthesis, and deployment via Vivado.
- Conducted packet analysis** between PC and FPGA board using loopback via Wireshark for effective debugging.
- Ported firmware** from KC705 with Kintex7 to AXKU040 with Kintex Ultrascale FPGA, adapting SGMII to RGMII for MAC-PHY Ethernet communication.

Undergraduate Researcher || HEAVY ION PHYSICS ANALYSIS

Sep 2024 – Jan 2025

PI: Dr. Prabhat Pujahari

EHEP Lab, IIT Madras, India

- Developed a custom gen fragment for $\Lambda_b \rightarrow \Lambda^0(p\pi^-) + J/\psi(\mu^+\mu^-)$ in PbPb collisions, **filter efficiency**: 1.3×10^{-6} .
- Performed full **Monte Carlo production** using the CMSSW framework, from generator-level raw data to miniAOD.
- Designed and deployed a CRAB3 configuration **optimized for large-scale production**, generating >1M rare events.

Undergraduate Researcher || COMPUTER VISION

Jan 2024 – May 2024

PI: Dr. Kaushik Mitra

Computational Imaging Lab, IIT Madras, India

- Developed GAN-based models with UNET architecture for flare artifact removal using restoration techniques.
- Utilized synthetically generated flare images from flare7kpp/Flickr24 datasets for training & testing models.
- Integrated Uformer, Restormer, NAFNet, Mamba blocks, and Retinexformer into the GAN-UNet codebase.
- Achieved PSNR (23.52) and SSIM (0.66) scores, while also enabling extension of the flare removal to videos.

NOTABLE PROJECTS

3D N-Body Gravitational Simulation

Jun 2023 – Mar 2024

AstroStellar Team Member, Centre for Innovation (CFI)

IIT Madras, India

- Generated initial galaxy conditions** using grid-based, angular, and radial Gaussian mass distributions.
- Solved Hamiltonian dynamics **using Leapfrog and Runge-Kutta** integrators for accurate numerical evolution.
- Parallelized code** to simulate more than **1 million** particles and visualized the simulation results in MATLAB.

Speaker and LED Composite System

Jan 2024 – May 2024

Course Project – EE2019 Analog Systems and Lab

IIT Madras, India

- Designed & demonstrated a **composite analog system** to synchronize light & sound with real-world functionality.
- Integrated DC-DC converter-based LED driver, bandpass filters, adder, peak detector, and Class-D audio amplifier.
- Analyzed and **resolved circuit non-idealities** to realize clean signal filtering and modulation.
- Simulated and validated** the system in LTSpice, showcasing analog design proficiency; awarded highest grade.

TEACHING

Foundations of Computational Physics, Course Instructor: Prof. Prabhat Pujahari

Aug 2025 – Nov 2025

- Teaching Assistant for an undergraduate course (~50 students) on numerical methods, simulation, and computational modeling with Python, C++, and ROOT, along with introductory modules on machine learning.
- Led weekly lab sessions, guided problem-solving, graded exams and assignments, and contributed to course design by creating projects/exam questions and gave an introductory talk on machine learning applications to physics.

TECHNICAL SKILLS

Programming Languages: C/C++, Python, MATLAB, VHDL, Verilog, Pythia

Data Analysis Tools: ROOT, uproot, Keras, Scikit-Learn, Numpy, Scipy, Pandas, Matplotlib, Wireshark, NetworkX

Hardware Devices: Oscilloscopes, Multimeters, Function Generators, Op-amps, Xilinx FPGA, Optical CMM (Zeiss)

Computer-Aided Design: Vivado, KiCAD, LTSpice, Geant4

Productivity Softwares: MacOS, Windows, Linux (Ubuntu), Bash/zsh, Git (Gitlab & Github), VSCode, L^AT_EX

PUBLICATIONS

- [1] CMS Collaboration, "A novel track finding algorithm to identify b-hadrons in b-jets using FusionNet: a geometric deep learning model." [CMS-DP-2025-035](#), 2025.

Lead author of the performance note publicly available on the CERN Document Server (CDS).

- [2] P. Kumar on behalf of the CMS Collaboration, "B-hadron identification in b-jets using a novel deep learning technique in pp and PbPb collisions in CMS." the European Physical Society Conference on High Energy Physics (EPS-HEP 2025), Marseille, France, 2025. Submitted to Proceedings of Science (PoS).

POSTER PRESENTATIONS

- [1] P. Kumar on behalf of the CMS Collaboration, "B-hadron identification in b-jets using a novel deep learning technique in pp collisions in CMS." Poster presented at the European Committee for AI in Fundamental Physics (EuCAIF) Conference (106), Sardinia, Italy, 2025.
- [2] P. Kumar, S. Ghosh, P. Pujahari, M. Nguyen, "B-hadron identification in b-jets using a novel deep learning technique in pp and PbPb collisions in CMS." Poster presented at All India Research Scholars Summit (AIRSS), IIT Madras, India, 2025.
- [3] AstroStellar Team (incl. P. Kumar), "Galaxy Collision Simulation with Adaptive N-body Integration." Poster presented at CFI Open House 2024, IIT Madras, India.

AWARDS AND ACCOMPLISHMENTS

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| • Research Program for International Talents Fellowship , École Polytechnique
<i>Selected for a 3-month paid research internship at Laboratoire Leprince-Ringuet (LLR) in expt. high-energy physics.</i> | 2025 |
| • Young Research Fellowship (~ ₹100K), IIT Madras
<i>Selected as one of 17 Young Research Fellows from a pool of 300+ students for funded undergraduate research.</i> | 2024 |
| • Ranked Top 5 in Engineering Physics Department, IIT Madras | 2024 |
| • Ranked Top 5% nationally among 150,000 students in Joint Entrance Examination (JEE) Advanced, India | 2022 |
| • Ranked Top 1% nationally among 1 million students in Joint Entrance Examination (JEE) Mains, India | 2022 |

KEY COURSES TAKEN

Physics: Quantum Mechanics^A, Classical Dynamics, Mathematical Phys^A, High Energy Phys^A, Statistical Phys.

Mathematics: Multivariable Calculus, Algebraic Topology^A, Differential Equations, Combinatorics^A.

Electrical: Analog & Digital Systems, Analog & Digital Signal Processing, Circuits and Networks, Solid State Devices.

Computational: Modern Scientific Computing, Computer Architecture^{A,†}.

VOLUNTEERING AND EXTRACURRICULARS

Saathi – Undergraduate Student Mentor

Jul 2024 – Present

- Guiding 4 college junior undergraduate students, helping them navigate from school to college and hostel life.

Shiksha Prayas – Student Mentor

Jan 2023 – Mar 2024

- Mentored two underprivileged high school students from Haryana, guiding them in their academic and personal development, leading to significant improvement and satisfactory board exam results.

^A: Graduate Level Course, [†]: Ongoing